

Name: _____

/ 25

30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. Consider the curves $y = x^2$, $y = -x + 2$, and $y = 0$.

(a) (4 points) Sketch the region bounded by these curves. Be sure to clearly label the curves and any important points. Show any relevant work.

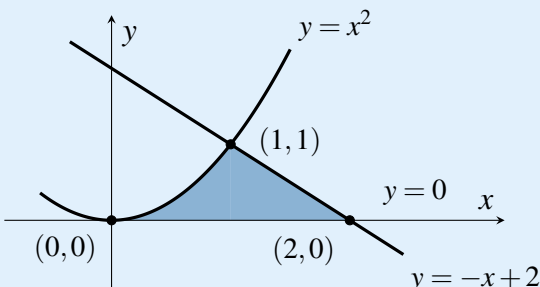
Solution

First find where the curves meet.

Parabola and line: $x^2 = -x + 2 \implies x^2 + x - 2 = 0 \implies (x+2)(x-1) = 0$, so $x = -2$ or $x = 1$. Only $x = 1$ borders our region, giving the point $(1, 1)$.

Parabola and x -axis: $x^2 = 0$ at the origin, $(0, 0)$.

Line and x -axis: $-x + 2 = 0$ at $x = 2$, giving $(2, 0)$.



(b) (6 points) Find the area of the region bounded by the curves.

Solution

Integrating in x , the top curve changes at $x = 1$, so the area splits into two pieces:

$$A = \int_0^1 x^2 dx + \int_1^2 (-x + 2) dx.$$

Evaluating each piece,

$$\int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3},$$

$$\int_1^2 (-x + 2) dx = \left[-\frac{x^2}{2} + 2x \right]_1^2 = (-2 + 4) - \left(-\frac{1}{2} + 2\right) = \frac{1}{2}.$$

Therefore

$$A = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$

Alternative (integrating in y). Here $\sqrt{y} \leq x \leq 2 - y$, so

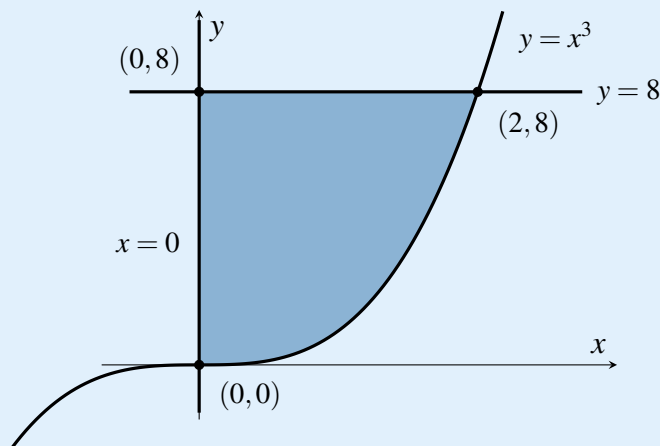
$$A = \int_0^1 \left[(2 - y) - \sqrt{y} \right] dy = \left[2y - \frac{y^2}{2} - \frac{2}{3} y^{3/2} \right]_0^1 = 2 - \frac{1}{2} - \frac{2}{3} = \frac{5}{6}. \quad \checkmark$$

2. Consider the region bounded by the curves $y = x^3$, $y = 8$, and $x = 0$.

- (a) (4 points) Sketch the region. Be sure to clearly label the curves and any important points. Show any relevant work.

Solution

The curves $y = x^3$ and $y = 8$ meet where $x^3 = 8$, i.e. $x = 2$, at the point $(2, 8)$. The line $x = 0$ meets $y = x^3$ at $(0, 0)$ and meets $y = 8$ at $(0, 8)$.



- (b) (6 points) Find the volume when the region is rotated around the y-axis.

Solution

Rotating about the y-axis, horizontal slices are perpendicular to the axis, so use **disks** in y. At height y the slice runs from $x = 0$ out to the cubic, $x = y^{1/3}$, so the radius is $R(y) = y^{1/3}$ and y runs from 0 to 8:

$$\begin{aligned} V &= \int_0^8 \pi (y^{1/3})^2 dy = \pi \int_0^8 y^{2/3} dy \\ &= \pi \left[\frac{3}{5} y^{5/3} \right]_0^8 = \frac{3\pi}{5} (8^{5/3}) = \frac{3\pi}{5} (32), \end{aligned}$$

so

$$V = \frac{96\pi}{5}.$$

- (c) (5 points) Set up, but **do not evaluate** an integral that computes the volume when the region is rotated around the x -axis.

Solution

Now the axis is horizontal, so slice vertically and use **washers** in x . At a given x in $[0, 2]$ the slice of the region stretches from $y = x^3$ up to $y = 8$; rotating it about the x -axis gives an annulus with outer radius $R = 8$ and inner radius $r = x^3$:

$$V = \int_0^2 \pi (8^2 - (x^3)^2) dx = \pi \int_0^2 (64 - x^6) dx$$